

A Synopsis Report
on
Hustlrzz — AI-Powered Mock Interview Coach

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For the degree of

Bachelor of Technology

in

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by

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Thank you all for your encouragement and belief in our project.

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ABSTRACT

Preparation for job interviews today depends largely on generic question banks, irregular peer practice, and feedback that arrives too late to influence the next attempt. Candidates also find it difficult to translate the evidence already present in their resume into a convincing narrative for a specific role. Hustlrzz is proposed as a web-based, AI-powered interview-preparation and career-coaching platform that brings resume analysis, role-specific interview practice, explainable coaching insight, and progress tracking together in a single secure workspace for students, graduates, career switchers, and working professionals.

The proposed system accepts a resume, a target job description, and optional company context, and returns a preparation pack containing role-relevant questions, answer direction, and role context. A Resume Analyzer processes PDF or DOCX files in memory to identify visible skills, likely gaps, a directional readiness score, and the single highest-value revision. A practice interview, conducted through typed or spoken responses over an authenticated WebSocket channel, produces adaptive follow-up questions and a structured session report. The platform will be implemented using Next.js and TypeScript for the interface, FastAPI for REST and WebSocket services with structured AI orchestration, and Supabase for authentication, PostgreSQL persistence, and row-level security, together with an optional pgvector-based retrieval layer restricted to candidate-approved material.

Every result is presented through a consistent insight pattern — status or score, plain-language takeaway, highest-priority action, and supporting evidence — so that a candidate understands not only how an attempt went but what to do next. Candidate data is protected through authenticated owner checks, row-level security, in-memory document processing, request limits, and daily quotas on high-cost analysis, while a provider-agnostic AI layer with fallback and timeout handling preserves availability when an external service is unreachable. The system is positioned deliberately as a coaching aid rather than an automated hiring or applicant-tracking-system decision maker, and its documented architecture leaves room for later extensions such as multilingual coaching and broader candidate-approved knowledge retrieval.

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1. INTRODUCTION

Hustlrzz is proposed as a web-based AI interview-preparation and career-coaching platform for students, graduates, career switchers, and working professionals. It will combine resume analysis, role-specific interview practice, simple coaching insights, and progress tracking in one secure workspace. The system is intended to replace generic question lists and irregular peer practice with tailored, repeatable preparation.

1.1 Problem Statement and Scope

Candidates often receive feedback too late, practice questions that do not match their target role, and struggle to convert a resume into a convincing interview narrative. The proposed solution will accept a resume, target job description, and optional company context; then produce a preparation pack, run typed or spoken practice, and identify one high-value next action. It will be a coaching aid, not an automated hiring or applicant-tracking-system decision maker.

Table 1: Problem-to-solution summary

Current challenge	Typical approach
Generic practice	Static question banks
Unclear feedback	Scores without context
Scattered tools	Separate utilities

1.2 Intended Users

The platform is intended for individual candidates, campus placement cells, career coaches, and training organisations. Individual users will practice privately; institutional users may use the platform to structure preparation activities, while candidate records remain owner-scoped and are not presented as hiring decisions.

2. OBJECTIVE

2.1 General Objective

To design and demonstrate a secure, accessible, and provider-resilient AI coaching platform that helps candidates prepare for interviews and improve their resumes for a chosen role.

2.2 Specific Objectives

- Generate role-specific questions, answer direction, and company/role context from a resume and job description.
- Analyse PDF or DOCX resumes to identify visible skills, likely gaps, and the highest-value revision.
- Provide a real-time typed or spoken interview using authenticated WebSocket communication and adaptive follow-up questions.
- Track presentation while the candidate answers — posture, gesture and eye contact — inside the browser, so camera frames are never sent to the server.
- Present every result as a simple insight: status or score, plain-language takeaway, highest-priority action, and supporting evidence.
- Protect candidate data through authenticated owner checks, row-level security, in-memory document processing, request limits, and browser security controls.
- Keep AI integration provider-agnostic so fallback handling can preserve availability when a provider is unavailable.

2.3 Success Metrics and Expected Evidence

Table 2: Success Metrics and Expected Evidence

Area	Measure	Expected evidence
Personalisation	Role relevance	Resume/JD-aligned question pack
Usability	Action clarity	One takeaway and next step per result

Reliability	Provider resilience	Fallback, timeout, and health checks
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2.4 Planned Deliverables and Boundaries

The deliverable will comprise a responsive candidate workspace, authenticated backend APIs, Resume Analyzer, practice interview flow, structured coaching report, and documented deployment configuration. It will deliberately exclude automated selection, unsupported medical/legal advice, unrestricted storage of raw candidate files, and organisation-wide sharing of private candidate material.

3. LITERATURE SURVEY AND SYSTEM DESIGN

Interview-preparation systems commonly offer question banks, video recordings, or generic feedback. Retrieval-augmented generation indicates that responses become more grounded when relevant source material is retrieved before generation; this supports an optional candidate-owned knowledge base built only from approved resume, portfolio, note, or previous-session material. Structured prompting supports a focused coaching flow: understand the role, assess evidence, and give a practical improvement.

Browser-based pose tracking and the Web Speech API show that lightweight presentation and speech features can run on modern devices without storing raw video. The research gap is the lack of a single, candidate-controlled workflow that connects resume review, role preparation, interview practice, and explainable feedback.

Table 3: Comparative analysis of similar existing platforms

Platform	Resume/JD Personalisation	Feedback Approach	Candidate Data Handling
Pramp (peer mock interviews)	None	Peer-driven, subjective	Session shared with peer partner
Big Interview	None	Rubric-based scripted review	Video stored on provider platform
Google Interview Warmup	None	Keyword/tone insight only	No persistent candidate profile
HireVue (employer-facing)	Employer-defined, not candidate-controlled	Automated hiring score	Data used for hiring decisions
Hustlrzz (Proposed)	Resume + JD + company context	Evidence-based, one priority action	Owner-scoped; raw files not retained

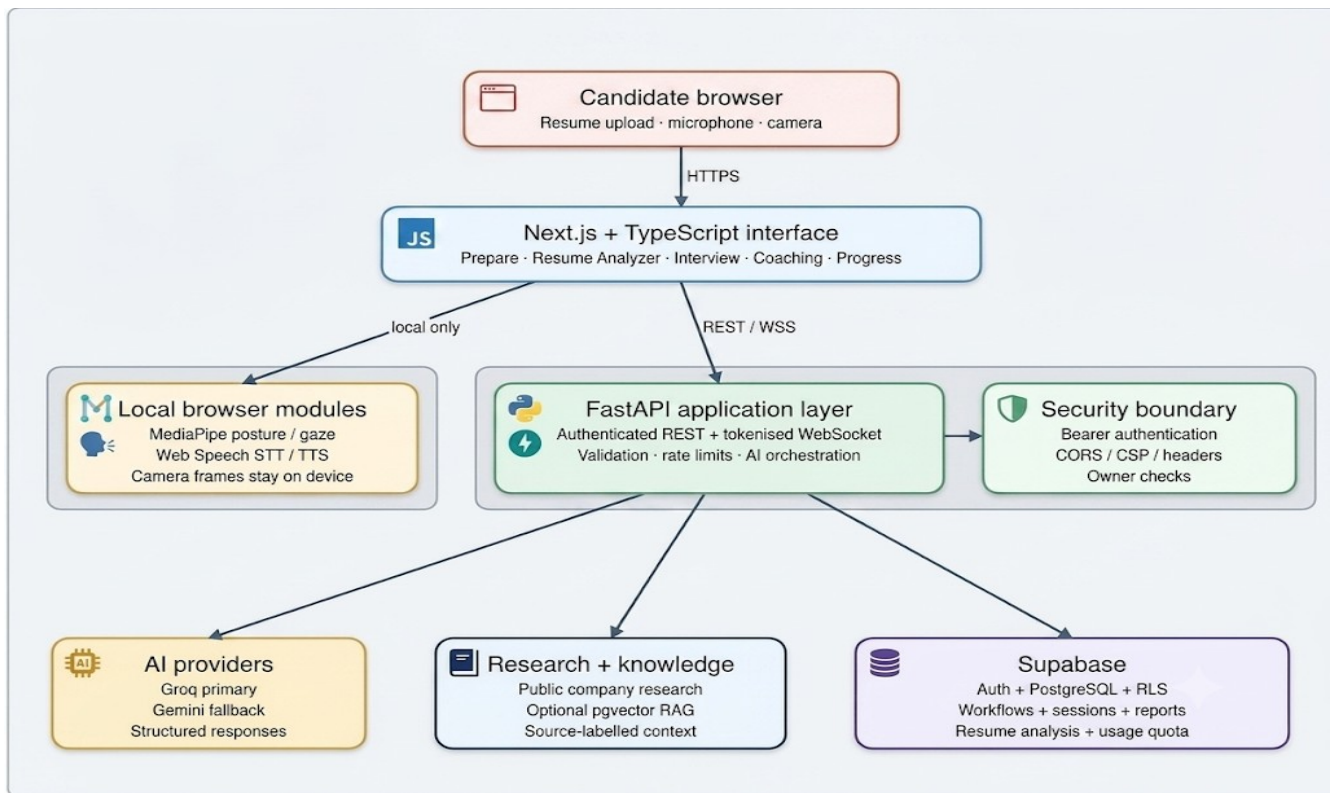


Figure 1: System Architecture of the AI-Powered Interview Platform

The architecture separates browser-local media processing from server-side AI orchestration and persistence. This separation reduces unnecessary transmission of camera data, keeps access controls at clear trust boundaries, and allows provider or knowledge components to evolve independently.

4. METHODOLOGY

4.1 Working Principle

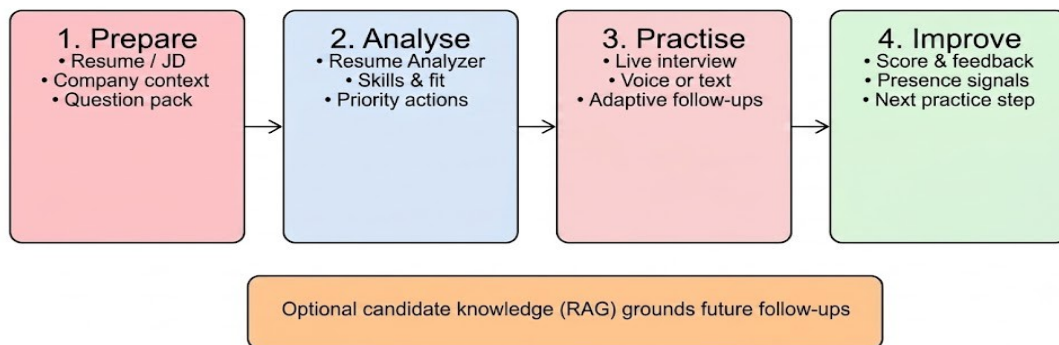


Figure 2: Four-Stage Candidate Workflow — Prepare, Analyse, Practice, Improve

- The candidate signs in and submits a resume, target job description, and optional context.
- The FastAPI application validates the request, extracts document content in memory, and coordinates structured AI analysis through a provider abstraction layer.
- The Next.js interface displays a preparation pack or Resume Analyzer insight, then opens a tokenised practice session through REST and WebSocket endpoints.
- The platform stores only owner-scoped structured results, workflows, sessions, and reports in Supabase; raw upload bytes are discarded after analysis.

4.2 Technical Approach

The frontend will use Next.js and TypeScript. FastAPI will provide REST and WebSocket APIs, structured AI orchestration, validation, and rate limits. Supabase will provide authentication, PostgreSQL persistence, and row-level security. Optional candidate retrieval will use embeddings and pgvector, with source-labelled context restricted to the same authenticated owner.

4.3 End-to-End Data Flow

Table 4: Proposed end-to-end data flow

Stage	Processing	Stored result
Input	Validate resume/JD; extract in memory	No raw upload retained
Analysis	AI orchestration and fallback	Structured workflow/analysis
Practice	Authenticated REST/WebSocket session	Transcript and report per owner
Coaching	Evidence-based insight generation	Priority action and progress

Quality will be checked through validation of inputs, structured-response parsing, authenticated ownership checks, actionable report review, and failure-path testing for provider timeouts or rate-limit responses. The interface will retain a consistent insight pattern so candidates can understand what happened and what to do next.

5. PROPOSED MODULES AND REQUIREMENTS

5.1 Core Functional Modules

Table 5: Proposed functional modules

Module	Candidate input	Planned output
Prepare	Resume, job description, company	Questions, answer direction, role context
Resume Analyzer	PDF/DOCX resume	Skills, gaps, directional score, revision
Interview	Typed/spoken answers	Adaptive follow-ups and session report
Coaching	Session/resume evidence	Takeaway, priority action, next practice
Knowledge base	Resume, portfolio, notes, session report, company research	Indexed chunks with similarity-scored search
Coaching workspaces	Session and resume evidence	Role fit, company playbooks, offer negotiation, practice room
Assessment	Timed multi-round responses	Graded rounds and a readiness report
Settings — provider key	Optional candidate provider credential	Encrypted at rest, shown only as a four-character hint
Account data rights	Authenticated owner	Full data export, or irreversible erasure of all records

5.2 Hardware Requirement/Specification

Table 6: Hardware requirement/specification

Component	Minimum Specification
Processor	Intel Core i5 or equivalent
RAM	8 GB minimum (16 GB recommended)
Storage	5 GB free disk space
Internet	Required for AI services and deployment

5.3 Software Requirement/Specification

Table 7: Software requirement/specification

Component	Specification
Operating System	Windows 10/11, Linux, or macOS
Frontend	Node.js 18+, Next.js, TypeScript
Backend	Python 3.11+, FastAPI
Database	Supabase PostgreSQL with Row-Level Security
Tools	VS Code, Git/GitHub, Postman, Docker

5.4 Security and Deployment Requirements

The proposed deployment will use environment-managed secrets, HTTPS, secure browser headers, allowed-origin controls, authenticated WebSocket sessions, owner-scoped access checks, daily usage quotas for high-cost analysis, and application health monitoring. A public launch will additionally require a shared work queue, distributed rate limiting, backup/recovery procedures, and provider-budget alerts.

5.5 Current Implementation Status

Work carried out so far covers the core candidate journey, from sign-in through to a written coaching result. The remaining modules have finished interfaces, but their backend behaviour is not yet wired. The table below records the position at this stage.

Table 8: Implementation status at the current milestone

Module	Status at this milestone
Prepare	Complete — resume or pasted text, job description, and optional company context produce a question pack with answer direction.
Resume Analyzer	Complete — PDF and DOCX analysed in memory; skills, gaps, directional score, and the single highest-value revision.
Interview	Complete — typed and spoken practice over an authenticated WebSocket, with adaptive follow-ups and a session report.
Coaching	Complete — evidence-based takeaway, priority action, and next practice step.
Knowledge base	In progress — source upload and search are implemented; retrieval quality is still being validated against real documents.
Coaching workspaces	UI built — backend behavior not yet wired.
Assessment	UI built — backend behavior not yet wired.

Settings and provider keys	UI built — backend behavior not yet wired.
Account data rights	UI built — backend behavior not yet wired.

The knowledge base is the one module that is partly built. Source upload and search are implemented, but retrieval quality is still being checked against real documents rather than only against test fixtures, so it is not being reported as complete.

One issue is worth recording because it changes how results are trusted. The knowledge base appeared to pass its automated tests, which supplied a stubbed embedding service. When it was exercised through the running application instead, the configured embedding model had been withdrawn by its provider and every request failed. The tests had therefore confirmed the code path, not the feature. Retrieval is now being re-validated against a live provider before the module is reported as complete.

The remaining planned modules, together with capacity planning, backup and recovery procedures, and evaluation with real users, form the next stage of work. Until the status above changes, none of them should be treated as available.

6. EXPECTED OUTCOMES AND APPLICATION

6.1 Expected Outcomes

- A working prototype where a candidate can prepare for a target role, analyse a resume, practice an interview, and receive focused coaching.
- A Resume Analyzer that accepts PDF/DOCX files, identifies visible and missing skills, provides a directional readiness score, and prioritises the next revision.
- Secure candidate isolation through authenticated access, owner checks, Supabase row-level security, rate limits, and controlled WebSocket sessions.
- Provider fallback, request timeouts, quotas for high-cost analysis, and health monitoring to support dependable operation.
- A documented, extensible architecture suitable for future additions such as multilingual coaching, shared AI work queues, and broader candidate-approved knowledge retrieval.

6.2 Responsible Use and Future Readiness

Table 9: Responsible-use design controls

Design control	Purpose
Owner checks + RLS	Keep each candidate's records isolated

In-memory file processing	Avoid retaining raw resume uploads
Rate limits + quotas	Control high-cost AI actions
Fallback + timeouts	Reduce disruption during provider failure

6.3 Application

Hustlrzz can support any job seeker who wants structured career preparation: students applying for internships or placements, graduates entering their first role, professionals changing careers, and candidates preparing for leadership or salary-negotiation conversations. It can also support placement cells, career-coaching programmes, and training organisations as a guided practice platform. Before a wider public release, the proposed system will require production capacity planning, shared queueing, distributed rate limiting, provider-budget controls, privacy review, and user evaluation.

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